NAMAA Community at the BAREC Shared Task 2026: Ordinal-Aware Training for Fine-Grained Arabic Readability Assessment Beyond Architectural Scale

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Abstract

Fine-grained Arabic readability assessment is challenging due to the language's rich morphology and the ordered nature of readability labels. We investigate this problem in the sentence-level Strict Track of the BAREC Shared Task 2026, which uses a 19-level readability scale. We evaluate six controlled configurations spanning engineered linguistic and morphological features, TF-IDF/SVD lexical representations, contextual transformer encoders, multi-encoder fusion, output-level ensembling, and ordinal-aware training, followed by a final refinement using a single EmbeddingGemma encoder. Contextual transformer representations provide the largest improvement over classical feature-based baselines, while multi-encoder fusion and voting yield smaller additional gains. In contrast, word-repetition and adjacent-word-swap augmentation degrades performance, indicating that these perturbations do not reliably preserve readability labels. Our strongest configuration combines EmbeddingGemma with Gaussian soft labels, KL-divergence training, expected-value decoding, class weighting, and best-checkpoint selection, achieving 84.0 QWK on the test set and 82.4 QWK on the official blind test set. Overall, the results suggest that explicitly accounting for the ordinal structure of readability labels is a promising alternative to simply increasing architectural or ensemble complexity. We further summarize the resulting design insights into a practical framework for Arabic ordinal readability modeling.

1 Introduction

Automatic readability assessment estimates the linguistic difficulty of a text, with applications in education, language learning, and accessibility (Liberato et al., 2024). Early systems relied on manually engineered surface features such as lexical statistics and sentence length (Uçar et al., 2024), which capture limited contextual, syntactic, or semantic information—especially for morphologically rich languages like Arabic (Cavalli-Sforza et al., 2018).

Recent work has shifted from Arabic-specific readability indices toward supervised and pretrained language-model approaches (Hazim et al., 2022). This shift was enabled by large annotated resources, notably the Balanced Arabic Readability Evaluation Corpus (BAREC) (Elmadani et al., 2025c), which provides sentence-level annotations across 19 ordered readability levels.

The BAREC Shared Task 2025 (Elmadani et al., 2025a) established strong baselines for this problem, with the best system reaching a Quadratic Weighted Kappa (QWK) of 87.5 using pretrained Arabic transformers, morphological processing, and ensembling (Elmadani et al., 2025b; Basem et al., 2025). However, since these systems combine multiple components, the individual contribution of each representation strategy remains unclear.

This work investigates neural representation strategies for sentence-level Arabic readability classification in the **BAREC Shared Task 2026** (Elmadani et al., 2026), focusing on contextual representations and attention-based aggregation under the 19-level ordered classification setting, with the goal of isolating which representations most effectively support readability prediction.

2 Background

Arabic readability research has progressed from manually designed formulas to supervised learning and large-scale neural benchmarks. Early feature-based approaches include Al-Khalifa and Al-Ajlan's readability predictor (Al-Khalifa and Al-Ajlan, 2010), the Automatic Arabic Readability Index (AARI) (Al-Tamimi et al., 2014), and OSMAN (El-Haj and Rayson, 2016), which incorporated diacritization and morphological complexity. These works established the value of Arabic-specific fea-

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tures but relied heavily on manual engineering.

Later studies framed readability as supervised classification using lexical, discourse, and syntactic features (Forsyth, 2014). Cavalli-Sforza et al. (Cavalli-Sforza et al., 2018) surveyed the field, highlighting challenges from Arabic morphology and limited annotated data. Complementary lexical resources, including a leveled readability lexicon (Al Khalil et al., 2020) and an associated readability-leveled thesaurus (Jiang et al., 2020), were also developed.

More recent efforts focus on dedicated readability datasets: SAMER (Alhafni et al., 2025) (book- and word-level annotations), DARES (El-Haj et al., 2024) (13,335 units from school materials), and the multilingual ReadMe++ (Naous et al., 2024). These resources mark the transition toward data-driven, pretrained-model-based readability assessment.

The **BAREC corpus** (Elmadani et al., 2025c) represents a major advance, comprising 69,441 sentences (over one million words) annotated across 19 fine-grained levels spanning kindergarten through higher education, following the Taha-Thomure scheme and dedicated annotation guidelines (Habash et al., 2025). Because labels are ordered, prediction errors between adjacent levels are less severe than between distant ones, distinguishing this task from standard categorical classification (for a large statistic study of the dataset look Appendix A).

The **BAREC Shared Task 2025**, with 22 teams from 12 countries, offered the first large-scale comparative evaluation under this framework, with the top system achieving 87.5 QWK (Elmadani et al., 2025b). Leading systems combined pretrained Arabic transformers, morphological features, and ensembling; the winning !MSA system, for instance, integrated four transformer models with CAMeL Tools morphological tokenization and prediction correction (Basem et al., 2025). As these systems combine multiple components, the individual contribution of each representation remains unclear.

The **BAREC Shared Task 2026** retains this 19-level framework, evaluated primarily via QWK and accuracy. Motivated by the limitations of surface-level features and the unresolved contribution of individual neural representations, we focus on contextual sentence representations and attention-based modeling for fine-grained Arabic readability prediction.

3 System Overview

We evaluate a sequence of six increasingly expressive systems (A–F) for the sentence-level Strict Track of the BAREC Shared Task 2026, progressing from manually engineered linguistic features to lexical (TF-IDF/SVD) representations, single- and multi-encoder transformer models, and ordinal-aware training with data augmentation. This progression allows us to isolate the contribution of each representation type to fine-grained readability prediction. Figure 1 summarizes the progression from the controlled Systems A–F to the final ordinal-aware EmbeddingGemma submission.

4 Experimental Setup

4.1 Dataset and Preprocessing

We use the official `CAMeL-Lab/BAREC-Shared-Task-2026-sent` dataset, consisting of ∼54.8K training, 7.3K validation, and 7.3K test sentences, each labeled on a 19-level ordinal readability scale. Predictions are additionally evaluated after mapping to the official 7-, 5-, and 3-level groupings. Final shared-task performance is evaluated separately on the official blind test set.

Under the Strict Track, all models are trained using only the BAREC training data, without external lexical resources. For System A, linguistic and morphological features are extracted using regex-based indicators and Stanza's Arabic UD pipeline (tokenization, POS tagging, dependency parsing); for System B, TF-IDF representations use word uni/bigrams and character 3–5-grams (`char_wb`), reduced via `TruncatedSVD` (fit on training data only) to a combined dimensionality of 192. For AraBERTv2 (Antoun et al., 2020), official Farasa-based (Abdelali et al., 2016) preprocessing is applied.

4.2 Model Architecture

System A trains a LightGBM regressor (MAE objective, 5-fold CV) on ∼100–200 engineered features (length/statistical, word-frequency, morphological, and POS/dependency features), pruned via cumulative gain importance and pairwise-correlation filtering ($|r| > 0.95$). **System B** concatenates the pruned linguistic features with the TF-IDF/SVD lexical representation, trained with the same LightGBM pipeline. **System C** fuses a frozen transformer sentence embedding (mean-pooled MARBERTv2 (Abdul-Mageed et al., 2021), with

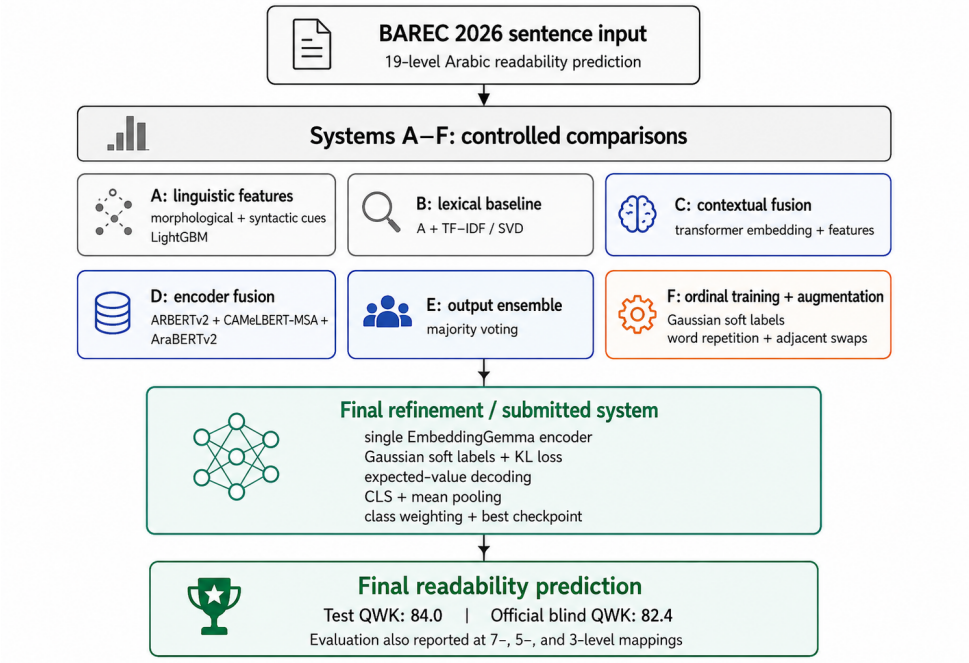


Figure 1: Overview of the BAREC 2026 system progression. Systems A–F provide controlled comparisons from linguistic and lexical baselines to contextual, multi-encoder, ensemble, and ordinal-aware configurations.

a staged variant using Qwen2.5-0.5B (Qwen et al., 2025)) with the linguistic and lexical branches via per-branch MLP projections (128-d) concatenated before a regression head. **System D** jointly fuses three pretrained Arabic encoders (ARBERTv2 (Abdul-Mageed et al., 2021), CAMeLBERT-MSA (Inoue et al., 2021), AraBERTv2), each tokenized separately, projected from 768 to 512 dimensions, concatenated (1536-d), and passed to a shared classification head over 19 levels. **System E** uses the same three encoders as System D but combines independent per-encoder predictions at the output level via soft-probability averaging or hard voting. **System F** extends the encoder-fusion setting with train-only data augmentation (retaining original examples and adding perturbed copies for classes representing $<1\%$ of training data via word repetition and adjacent-word swaps) and ordinal-aware training: Gaussian soft labels with an ordinal KL-divergence objective, expected-value (soft-argmax) decoding, and CLS+mean-pooled representation concatenation.

4.3 Training Configuration

Systems A and B use continuous regression outputs converted to ordinal levels via an `OptimizedRounder` fitted by maximizing QWK with Nelder–Mead optimization. Systems C–F are trained with 5-fold cross-validation (C) or standard supervised training (D–F), using `SmoothL1Loss` (C) or cross-entropy/KL-divergence objectives (D–F), with checkpoints selected by validation-set QWK after each epoch. System F applies differential learning rates ($1e{-5}$ for encoders, $5e{-5}$ for the classification head) and inverse-frequency class weighting based on the post-augmentation training distribution. For all neural systems, the best development-QWK checkpoint is reloaded prior to final evaluation on the validation and test sets.

System	Acc19	Acc±1	MAE↓	QWK
A	19.98	51.96	1.726	70.60
B	22.33	56.92	1.595	73.67
C	54.87	70.59	1.101	81.12
F	54.12	69.83	1.142	79.15
D	55.38	71.16	1.083	81.50
E	**56.09**	**72.03**	**1.068**	**81.76**

Table 1: Test-set results (%) on the BAREC 2026 sentence-level Strict Track. Higher is better except MAE.

5 Results

5.1 Experimental Configurations

We report exact 19-level accuracy (Acc19), accuracy within one level (Acc±1), mean absolute error (MAE), Quadratic Weighted Kappa (QWK, the pri-

System	Acc19	Acc±1	MAE↓	QWK
A	19.49	50.75	1.788	70.29
B	22.26	55.83	1.646	73.44
C	54.23	69.75	1.155	80.16
F	52.95	69.38	1.179	79.79
D	53.75	70.08	1.167	80.11
E	**54.92**	**70.57**	**1.133**	**80.91**

Table 2: Validation-set results on the BAREC 2026 sentence-level Strict Track. Acc19, Acc±1, and QWK are reported in percentage points; higher is better for all metrics except MAE. System E achieves the best overall validation performance.

mary metric), and accuracy at the coarser 7-, 5-, and 3-level groupings (Acc7, Acc5, Acc3), following the official shared-task protocol.

5.2 Main Results

Table 1 reports test-set results; Table 2 reports the corresponding validation-set results.

System A obtains a QWK of 70.60% from engineered features alone, though exact accuracy remains low (19.98%), reflecting the difficulty of distinguishing all 19 levels from surface statistics alone. Adding TF-IDF/SVD lexical features (System B) improves QWK to 73.67% and accuracy to 22.33%. Introducing a contextual transformer representation (System C) yields the largest single gain, raising QWK to 81.12% and accuracy to 54.87%, indicating that contextual representations capture readability signal not reflected in engineered or sparse lexical features. Multi-encoder fusion provides a further, smaller improvement: System D reaches 81.50% QWK, and the output-level voting ensemble (System E) achieves the best overall performance (81.76% QWK, 56.09% accuracy), also obtaining the highest Acc7 (65.95%), Acc5 (69.50%), and Acc3 (74.95%). Voting (System E) modestly outperforms feature-level fusion (System D) by 0.26 QWK points and 0.71 accuracy points. System F, incorporating data augmentation and ordinal-aware training, underperforms the non-augmented fusion systems (79.15% QWK), suggesting that word-repetition and adjacent-word-swap perturbations do not reliably preserve readability labels. Validation-set results (Table 2) follow the same trend, with System E again achieving the best performance (80.91% QWK) across all label granularities.

Based on these results, the three-encoder ensemble (System E) was our strongest configuration among Systems A-F and served as the baseline for the final refinement described below.

As a final experiment, we apply the ordinal-aware refinements from System F to a single-branch **EmbeddingGemma** encoder (Vera et al., 2025): (1) an ordinal KL-divergence loss over Gaussian soft labels replacing cross-entropy; (2) expected-value (soft-argmax) decoding at test and blind-test time; (3) concatenated CLS and mean-pooled representations; (4) differential learning rates ($1e{-5}$ encoder, $5e{-5}$ head); (5) reloading the best development-QWK checkpoint rather than last-epoch weights; and (6) inverse-frequency class weighting.

This configuration reaches **84.0 QWK** on the test set and **82.4 QWK** on the blind test set, surpassing all other evaluated systems, including the three-encoder voting ensemble (System E). Since the backbone, objective, pooling, decoding, class weighting, and checkpoint selection all change jointly, we do not attribute this gain to any single component.

5.3 Analysis and Discussion

Contextual representations yield the largest gain over feature-based baselines, while added multi-encoder complexity contributes comparatively little. Our largest further gain instead comes from the ordinal-aware EmbeddingGemma configuration, suggesting that modeling the ordinal label structure deserves at least as much attention as scaling model or ensemble complexity. Motivated by this, we distilled a practical framework for ordinal readability prediction from prior BAREC and Arabic readability work alongside our own results (Appendix B), intended as a reference for future work on this and similar ordinal text-classification tasks.

6 Conclusion

We compared feature-based, contextual, multi-encoder, ensemble, and ordinal-aware approaches for fine-grained Arabic readability assessment in BAREC 2026. Contextual representations gave the largest improvement over classical baselines; multi-encoder fusion and ensembling added smaller gains, while simple text augmentation hurt performance. Our strongest configuration—a single EmbeddingGemma encoder with ordinal-aware training and decoding—reached 84.0 QWK on the test set and 82.4 QWK on the blind test set, outperform-

ing larger ensemble systems.

Acknowledgments

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A Appendix A

This appendix provides a comprehensive visual analysis of the dataset and its learned representations. We first examine the distribution of linguistic and readability-related characteristics across the readability classes, including sentence-level class distribution, average word length, maximum word length, sentence word count, and the relationship between sentence length and readability. We additionally analyze the correlations among the extracted linguistic variables using a Spearman correlation matrix. These analyses provide an overview of the structural characteristics of the dataset and help assess whether the readability classes exhibit distinguishable linguistic patterns.

Figure 2 presents the distribution of sentences across the readability classes. The figure provides an overview of the class balance and highlights any potential differences in the number of instances available for each readability level.

Figure 3 shows the average word length across readability classes. This analysis investigates whether lexical length varies systematically with readability level.

Figure 4 illustrates the distribution of maximum word length across the readability classes. Maximum word length provides an additional indicator of lexical complexity and may capture the presence of morphologically or lexically complex words.

Figure 5 presents the Spearman correlation matrix between the extracted linguistic features. This analysis is used to examine monotonic relationships between variables and to identify potentially redundant or strongly associated features.

Figure 6 shows the distribution of the number of words per sentence across the readability classes. Sentence length is a commonly used indicator of textual complexity, and its distribution provides insight into how sentence-level length varies across the different readability levels.

Figure 7 further examines the relationship between sentence length and readability. The visualization illustrates whether longer sentences tend to correspond to higher or lower readability levels and whether sentence length alone provides a clear separation between classes.

A.1 Embedding Space Visualization

To complement the descriptive analysis, we further investigate whether readability classes exhibit meaningful structure in contextual representation

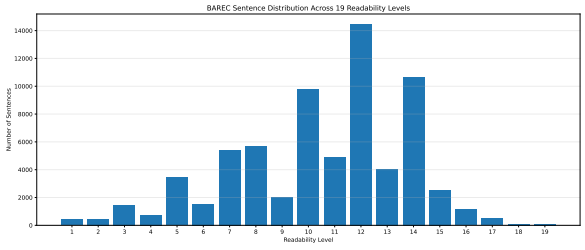


Figure 2: Distribution of sentences across the readability classes.

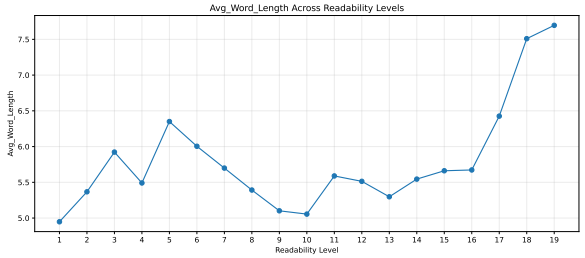


Figure 3: Average word length across readability classes.

space. We extract sentence-level embeddings using two complementary pretrained language models: ARBERTv2 and EmbeddingGemma-300M. The resulting representations are projected into two- and three-dimensional spaces using both t-SNE and UMAP.

Figures 8–15 visualize these representations. The 2D and 3D projections provide complementary views of the geometry of the embedding space, while the use of both UMAP and t-SNE allows us to examine whether local and global neighborhood structures are consistently reflected across dimensionality-reduction methods.

For ARBERTv2, Figures 8 and 9 show the two-dimensional UMAP and t-SNE projections, respectively. The corresponding three-dimensional projections are presented in Figures 10 and 11. Similarly, Figures 12 and 13 present the two-dimensional projections obtained from EmbeddingGemma-300M, while Figures 14 and 15 provide their three-dimensional counterparts.

These visualizations are intended as an exploratory analysis rather than as quantitative evidence of separability. In particular, dimensionality-reduction methods can distort distances and neighborhood relationships when projecting high-dimensional representations into low-dimensional spaces. Nevertheless, consistent clustering or gradual transitions across multiple projection methods and embedding models can provide useful qualitative evidence regarding the extent to which readability information is encoded in the learned representations.

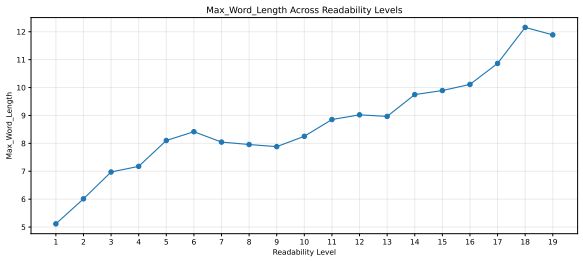


Figure 4: Maximum word length across readability classes.

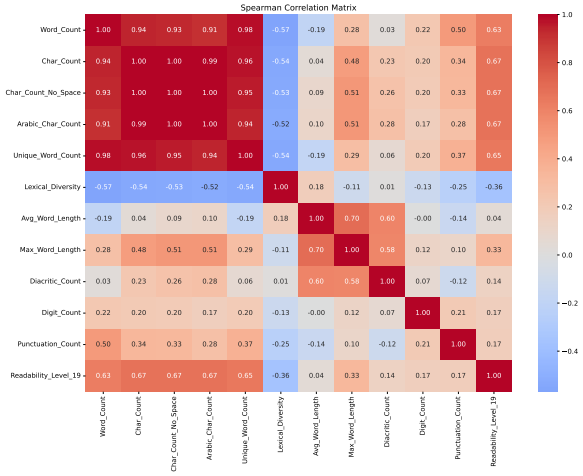


Figure 5: Spearman correlation matrix of the extracted linguistic features.

B Appendix B

Based on the experimental findings presented in this work, we propose a general six-stage framework for researchers and practitioners working on Arabic readability assessment. The framework is intended as a practical guide rather than a prescriptive pipeline, since the effectiveness of individual components may depend on the dataset, target metric, and evaluation setting.

B.1 Text Processing

The first stage concerns the preprocessing and representation of the input text. Experiments indicate that directly using the original text is a useful baseline; however, normalization and linguistic preprocessing can provide additional benefits. In particular, Arabic character normalization, stemming, and removal of noise can improve downstream performance in several experimental settings.

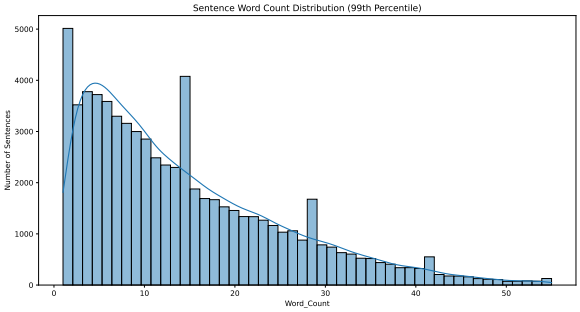


Figure 6: Distribution of the number of words per sentence across readability classes.

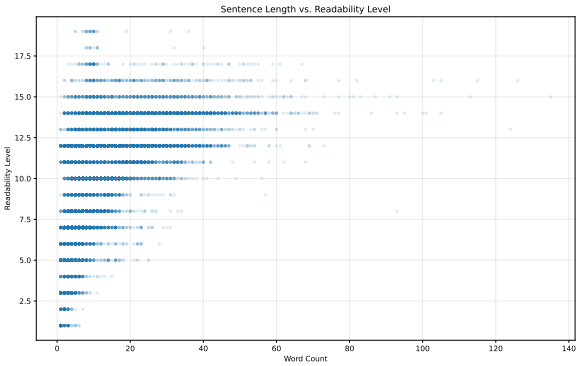


Figure 7: Relationship between sentence length and readability level.

Nevertheless, preprocessing should not be applied blindly. Each transformation can introduce artifacts or remove information that may be relevant to readability. Therefore, after applying normalization, stemming, or cleaning operations, the resulting text should be inspected for newly introduced noise, incorrect character transformations, or loss of linguistically meaningful information. Different preprocessing configurations should consequently be evaluated empirically rather than assuming that more aggressive preprocessing will always lead to better performance.

B.2 Sparse Representations and Classical Machine Learning

The second stage considers sparse lexical representations, particularly TF–IDF. TF–IDF provides a strong and computationally inexpensive baseline for readability assessment and can be combined with dimensionality-reduction techniques such as Principal Component Analysis (PCA) when appropriate.

The resulting representations can then be provided to conventional machine-learning algorithms, including Support Vector Machines (SVM), Random Forests (RF), Decision Trees (DT), and other suitable classifiers or regressors. Although these approaches do not explicitly model contextual semantics, they remain useful because they provide interpretable baselines and can reveal how much of the readability signal can be captured through lexical statistics alone.

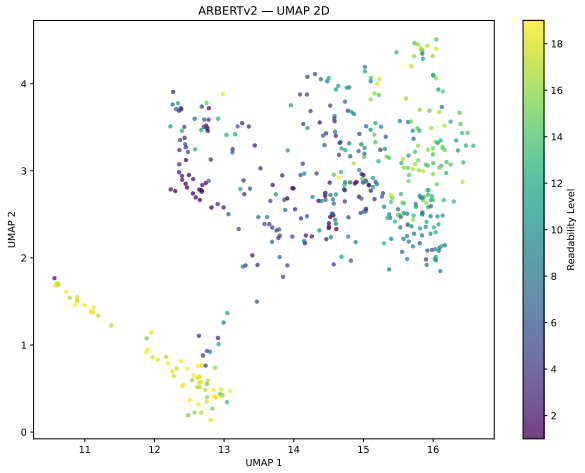


Figure 8: Two-dimensional UMAP projection of ARBERTv2 sentence representations, colored by readability class.

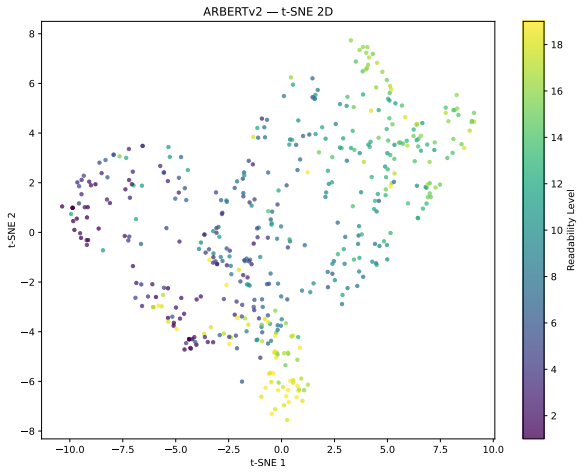


Figure 9: Two-dimensional t-SNE projection of ARBERTv2 sentence representations, colored by readability class.

B.3 Contextual Embeddings and Fine-Tuning

The third stage involves contextual neural representations. Encoder-based Arabic language models such as ARBERTv2 can be fine-tuned directly for the target readability task. In addition, newer embedding-oriented models, such as EmbeddingGemma-300M, showed promising behavior in our experiments and provide an alterna-

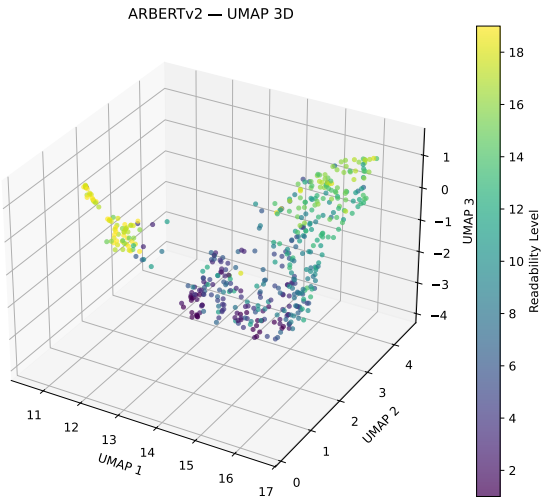


Figure 10: Three-dimensional UMAP projection of ARBERTv2 sentence representations, colored by readability class.

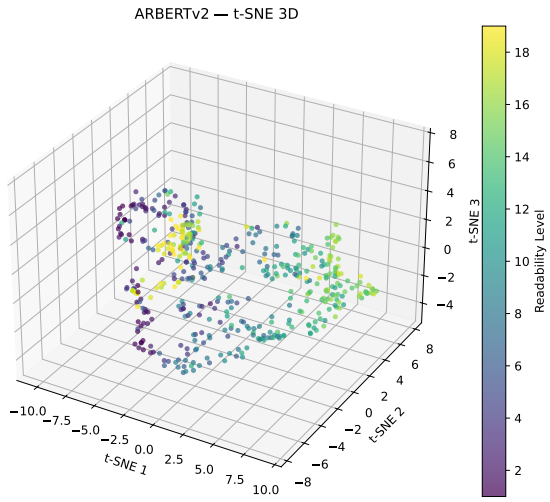


Figure 11: Three-dimensional t-SNE projection of ARBERTv2 sentence representations, colored by readability class.

tive representation space for downstream readability modeling.

Fine-tuning should be evaluated under the same data splits and evaluation protocol used for the classical baselines. This allows improvements to be attributed to the contextual representations rather than differences in preprocessing or experimental settings.

B.4 Objective Functions and Evaluation Metrics

The choice of training objective is particularly important for ordinal readability assessment. Readability levels have an inherent ordering, meaning that confusing adjacent levels is generally less severe than confusing distant levels. Consequently, conventional cross-entropy (CE) does not explicitly model the ordinal structure of the labels.

Across our experiments, ordinal-aware objectives generally produced better results when Quadratic Weighted Kappa (QWK) was the primary evaluation criterion. This suggests that incorporating the ordering of readability levels into the learning objective can be beneficial for this task. However, the behavior is not necessarily identical for all evaluation metrics. In particular, further experimentation is required to determine whether ordinal objectives consistently improve exact accuracy and ± 1 accuracy. Therefore, the choice of loss should ultimately be aligned with the target evaluation metric rather than optimized exclusively for one measure.

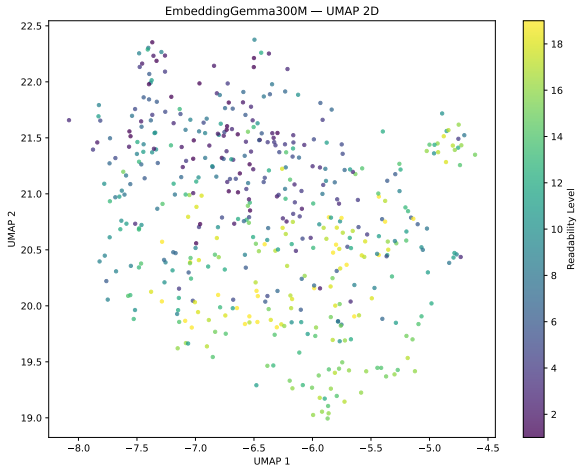


Figure 12: Two-dimensional UMAP projection of EmbeddingGemma-300M sentence representations, colored by readability class.

B.5 Feature Fusion and Linguistic Enrichment

A further direction is to combine contextual representations with complementary linguistic information. Instead of relying on a single embedding model, representations from multiple pretrained models can be fused to capture complementary aspects of the input text. For example, embeddings from different Arabic language models may encode different lexical, syntactic, or semantic properties.

The representation can also be enriched with explicitly extracted linguistic features, including lexical statistics, morphological complexity, part-of-speech information, syntactic properties, and dependency or constituency-based features. Such

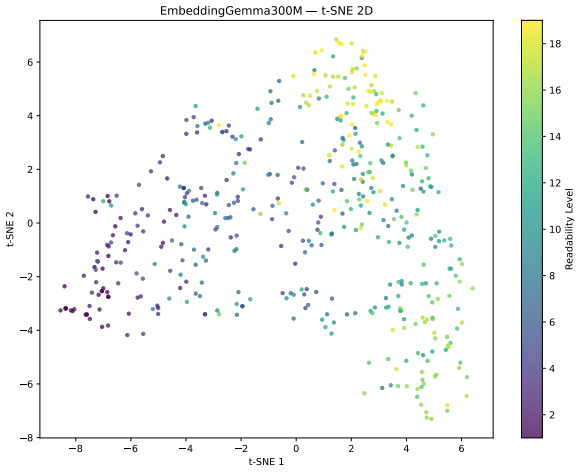


Figure 13: Two-dimensional t-SNE projection of EmbeddingGemma-300M sentence representations, colored by readability class.

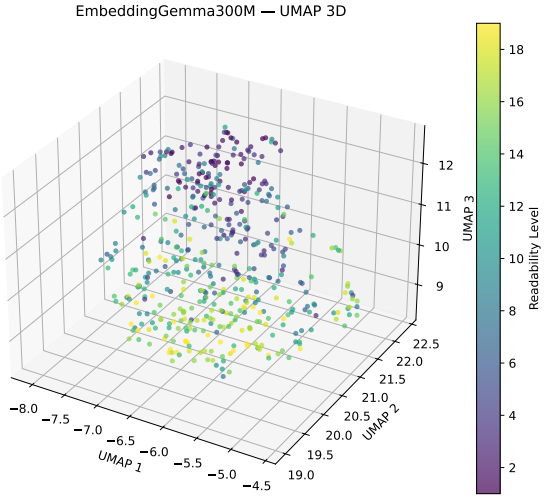


Figure 14: Three-dimensional UMAP projection of EmbeddingGemma-300M sentence representations, colored by readability class.

features may provide information that is not fully captured by the contextual representation, particularly for readability, where linguistic complexity is not necessarily equivalent to semantic similarity.

B.6 Ensemble and Hybrid Pipelines

Finally, ensemble approaches provide another promising direction. Our experiments suggest that combining predictions or representations from heterogeneous pipelines can improve robustness compared with relying on a single model. In particular, classical TF–IDF-based models, fine-tuned contextual models, embedding-based classifiers, and linguistically enriched models can provide complementary signals.

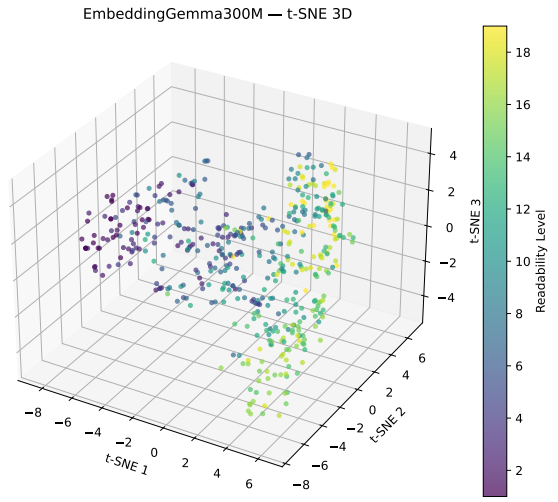


Figure 15: Three-dimensional t-SNE projection of EmbeddingGemma-300M sentence representations, colored by readability class.

A general high-level pipeline is therefore:

1. **Process the text:** evaluate the raw text and different Arabic normalization, stemming, and noise-removal configurations.

2. **Establish sparse baselines:** construct TF–IDF representations and evaluate classical machine-learning algorithms, optionally with dimensionality reduction.

3. **Learn contextual representations:** fine-tune Arabic pretrained models and evaluate newer embedding models such as EmbeddingGemma-300M.

4. **Model the ordinal objective:** investigate ordinal-aware losses when QWK is the primary metric, while separately monitoring accuracy and ± 1 accuracy.

5. **Enrich the representation:** investigate fusion of multiple embeddings with lexical, morphological, and syntactic features.

6. **Combine complementary systems:** evaluate ensemble or hybrid pipelines to exploit complementary strengths of different representations and modeling paradigms.

Overall, the framework emphasizes that Arabic readability assessment should not be treated as a single-model problem. Strong performance can arise from the interaction between appropriate text preprocessing, lexical representations, contextual

embeddings, ordinal-aware objectives, linguistic information, and ensemble strategies. The relative contribution of each component should, however, be established through controlled ablation experiments rather than assumed a priori.